# RAGUL JAYARAJ

GenAI Backend Engineer

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**LinkedIn:** www.linkedin.com/in/ragul315 | **GitHub:** github.com/ragul315 | **Portfolio:** ragul315.github.io/portfolio

## SUMMARY

Building enterprise **GenAI backends** at UST: orchestrating **asynchronous AI agents** via Temporal, securing LLM pipelines with **responsible-AI guardrails**, and engineering **fault-tolerant Python/FastAPI** services optimized with Redis caching.

## SKILLS

**GenAI & Agents:** AI Agents, Model Context Protocol (MCP), LLM response validation, RAG, LiteLLM
**Responsible AI:** AI governance, PII protection (Presidio), Content moderation, Azure AI Content Safety
**Backend:** Python, FastAPI, Pydantic, SQLAlchemy, REST APIs, Temporal
**Data & Caching:** PostgreSQL, MSSQL, MongoDB, MySQL, Redis
**Reliability:** Circuit breakers, Back-off retries, Caching, Performance tuning, Pytest
**Frontend & Tooling:** React, TypeScript, Docker, Git, Linux, Postman

## EXPERIENCE

**Software Engineer**, UST, Trivandrum, Kerala *Dec 2025 – Present*

- Implemented **responsible-AI guardrails** achieving **78% PII detection accuracy** via Presidio and a Factory-pattern content-moderation layer running on Azure AI Content Safety.
- Built **Temporal workflows** with custom retry policies for asynchronous backend pipelines and AI agents, reducing task execution failures by **45%**.
- Deployed **Redis caching** that cut database calls by **40%**; embedded circuit breakers and retries to protect workflows from downstream service failures.
- Enforced strict **LLM response validation** with Pydantic schemas, reducing downstream parsing crashes by **80%**.
- Resolved production bottlenecks by profiling backend services and optimizing slow SQL execution plans; refactored hardcoded SQL into a multi-database SQLAlchemy ORM, removing MSSQL lock-in.

**Embedded Software Intern**, Onwords Smart Solution, Pollachi, Tamil Nadu *Jun 2023*

- Delivered **IoT water tank automation software** streaming live water-level data over MQTT and automating pump control with **90% accuracy**, with remote monitoring and overflow protection; in use on the client side.
- Programmed ESP32 and ESP8266 microcontrollers in Embedded C for sub-second real-time device communication over MQTT.

## PROJECTS

**Finovo (team project)** github.com/ragul315/Finovo

- Architected an **async FastAPI and React microfinance platform** handling automated loan origination, EMI schedules, and Stripe payment gateway integration.
- Built a **zero-anomaly double-entry ledger** and cryptographic hash-chain audit log for tamper-evident administrative security.

**Student Task Management System** github.com/ragul315/Student-Task-Management-System

- Developed a secure role-based task platform featuring OAuth2 and JWT authentication, backed by comprehensive Pytest and Jest test suites.

## EDUCATION

**Bachelor of Engineering (B.E.) in Computer Science and Engineering** *2021 – 2025*
Dr. Mahalingam College of Engineering and Technology | CGPA: 8.0

## CERTIFICATIONS

Database and SQL (Infosys Springboard) | MySQL & Python Basic (HackerRank) | Python Elite+Silver (NPTEL)